Pluralis Agora

The DHT-based architecture behind the Pluralis-8B run

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Pluralis-8B at a glance

8B
parameters

7
pipeline stages

40 days
of training

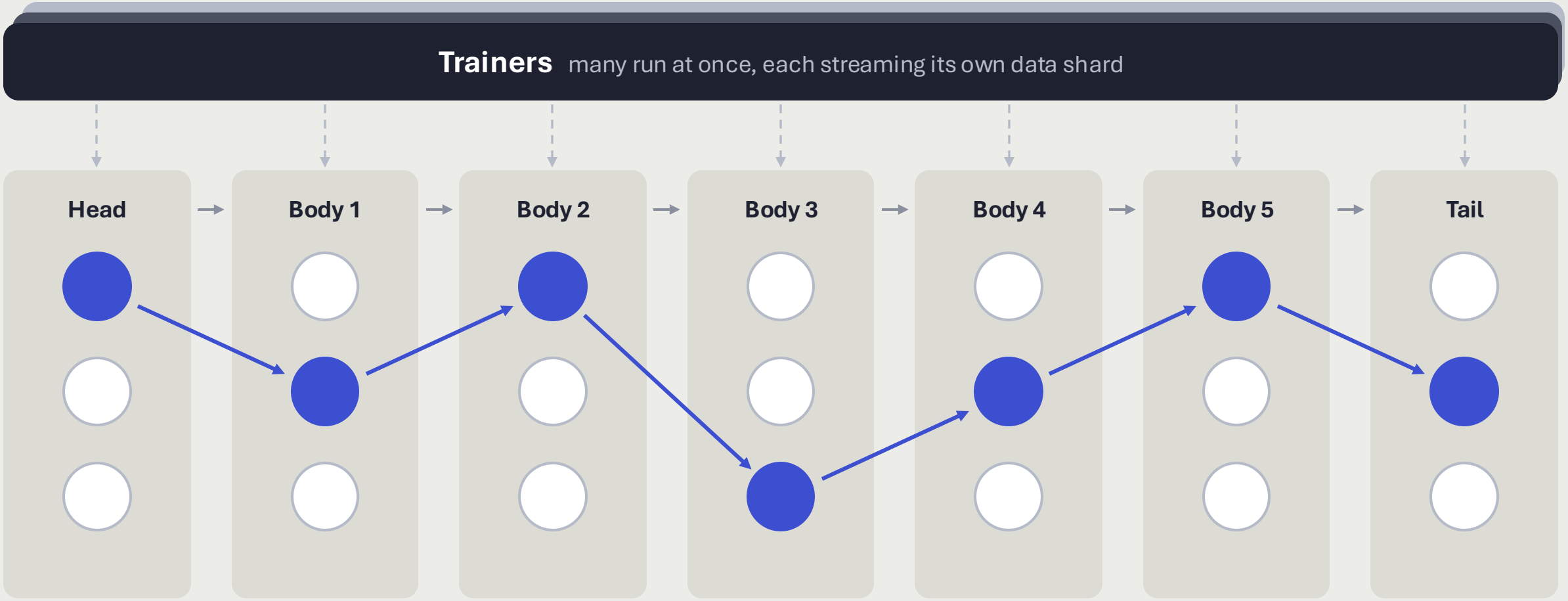
500B
tokens processed

330
contributor nodes

FineWeb-Edu · bf16 · 14 May → 23 Jun 2026

Paper-reported figures; the public dashboard shows 531.8B tokens over 51 days. Source: Agora paper §5.2–5.3; agora.pluralis.ai

Model split into Stages



Source: Agora paper, Figure 1 and §3.1 (arxiv.org/html/2607.13332v1#S3.SS1)

Contributers and Reducers

	Reducer node	Contributor node
Averaging	**Averages one weight slice, sends it back**	**Sends its slices, applies the averages**
If it drops out	Its slice keeps local values that round	Round finishes without it

Both roles are workers in the same stage. Source: Agora paper §3.3.1 (reducer role, fault tolerance), §5.1 (run setup)

AllReduce Operation

Hivemind

Every peer holds the full model

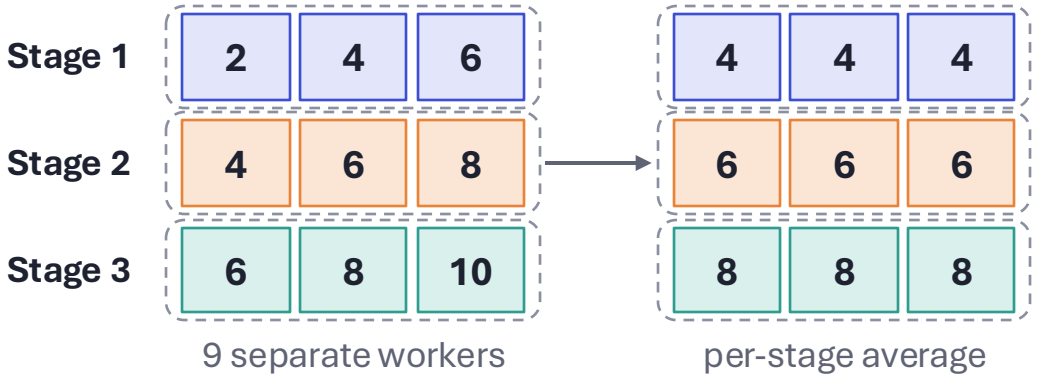


One average over all stages, on every peer

Every peer averages the whole model with all peers

Agora

Each worker holds one stage

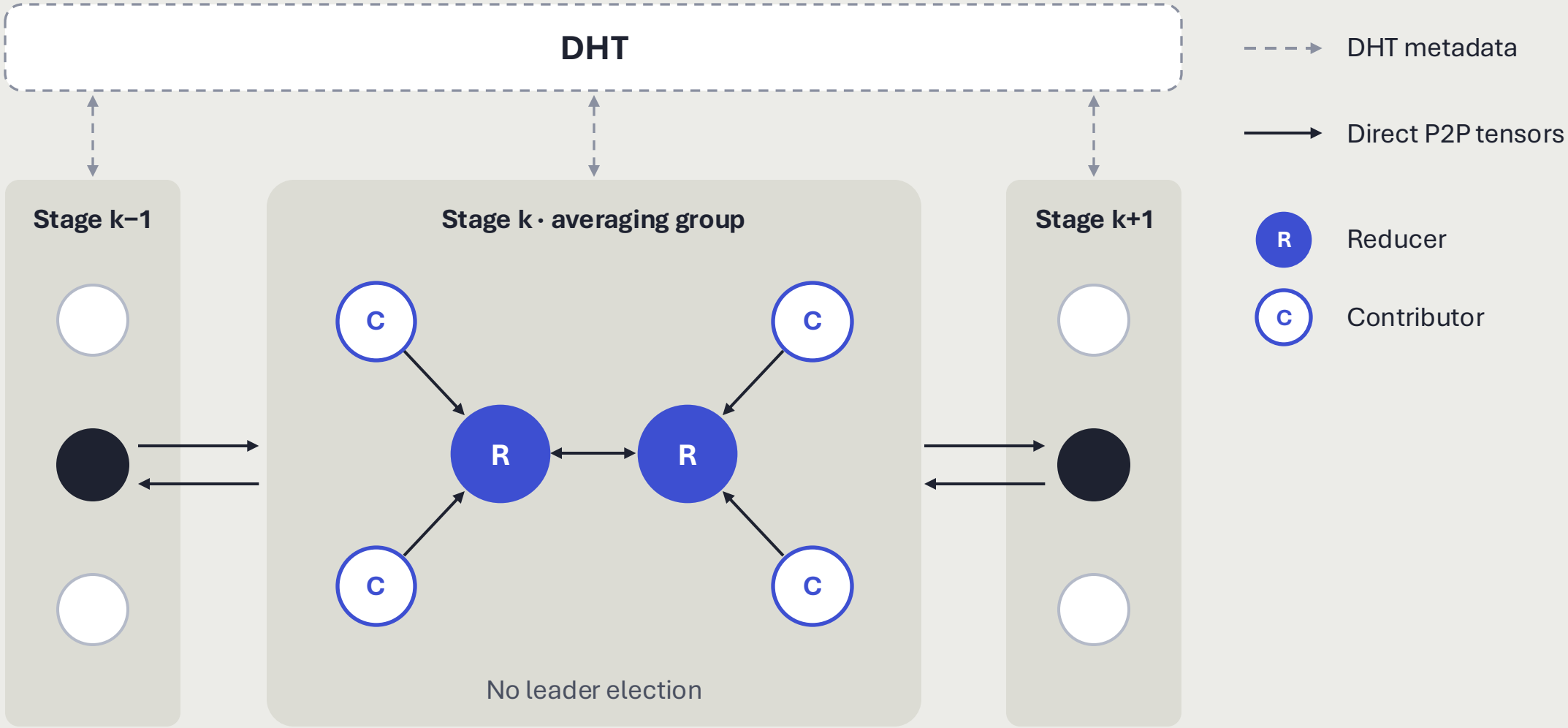


A separate average inside each stage

Workers average only with replicas of their own stage

Sources: Hivemind optimizer (hivemind/optim/optimizer.py:35); Agora paper §3.2 (arxiv.org/html/2607.13332v1#S3.SS2)

Averaging



Source: Agora paper, Algorithm 3 and §5.9.2 (arxiv.org/html/2607.13332v1#S5.SS9.SSS2)

Reducers average 5% of the weights every 20 steps

When and what

Every 20th stage step



step 1 — **step 20: average**

1 step = 2,048 sequences across the stage

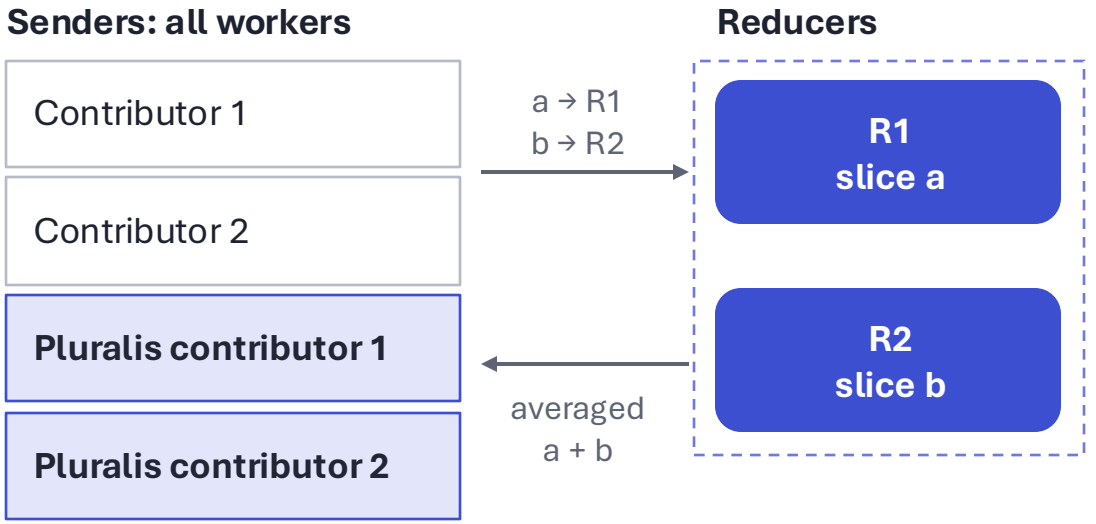
A rotating 5% of the weights



All weights covered every 400 steps

Training never pauses

Who averages



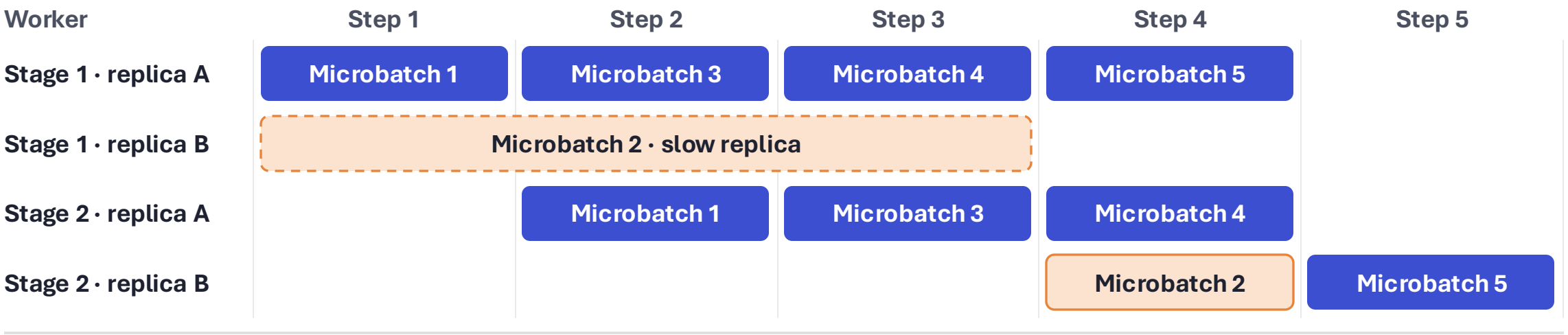
Both averaged slices return to all workers

Pluralis-8B values. Source: Agora paper §3.3.1, Algorithm 3

A slow replica delays only its own microbatch

Timeline with one slow replica

Stages 1 and 2, two replicas each



Trainers route new microbatches to fast replicas

Illustrative example. Source: Agora paper §3.2 (arxiv.org/html/2607.13332v1#S3.SS2)

Joining happens once; averaging repeats every round

One peer's view

1
Load config
Model + stage

2
Join the DHT
Bootstrap peers

3
Advertise the key
prefix + step

4
Build a roster
No leader

5
All-reduce
Peer-to-peer

Once, on join

Every round: every 20 steps, new key

Source: Agora paper §3.3.1, Algorithm 3, §5.9.2; step 1 settings from our harness

Agora is still centralized in four places

What Pluralis kept in this run

Admission

One Authorizer admits every node

Single gatekeeper

Head and tail

Contributors join body stages only

Head 8 · tail 16 nodes

Averaging

Only Pluralis nodes reduce

Contributors only send

Starting state

Pluralis ran the first 5,000 steps

Joiners load its saves

In the paper's words

“This run was not decentralized and required specific Pluralis nodes.”

Source: Agora paper §3.3 (training roles), §3.3.1 (reducer role), §3.4.3 (stage bottlenecks), §3.6.1 (Authorizer), §5.1 (run setup)

Contributors needed fast links and large GPUs; 669 joined and 607 left over 40 days

Admission requirements

≥ 200 Mbps
download
Median: 703 Mbps

≤ 80 ms
round-trip time
Median: 28 ms

≥ 24 GB
GPU memory

≥ 80 GB
system RAM

Membership over the 40-day run

669
joins
≈ 17 per day

607
departures
≈ 15 per day

63
peak concurrent contributors
cap: 60

Rejoins count as new events. Source: Agora paper §5.2–5.3 (admission thresholds, measured contributor network, membership events)

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Agora run results: Pluralis-8B telemetry

Matchmaking

5 s

median time to form a roster

0 s — 40 s window

Typical roster: 16 peers

DHT lookups

67 ms

median read (write: 68 ms)

99th percentile: 471 ms read, 1,110 ms write

Slow intervals: 0.5–1.1 s

Weight averaging

99.3%

of averaged weight data received

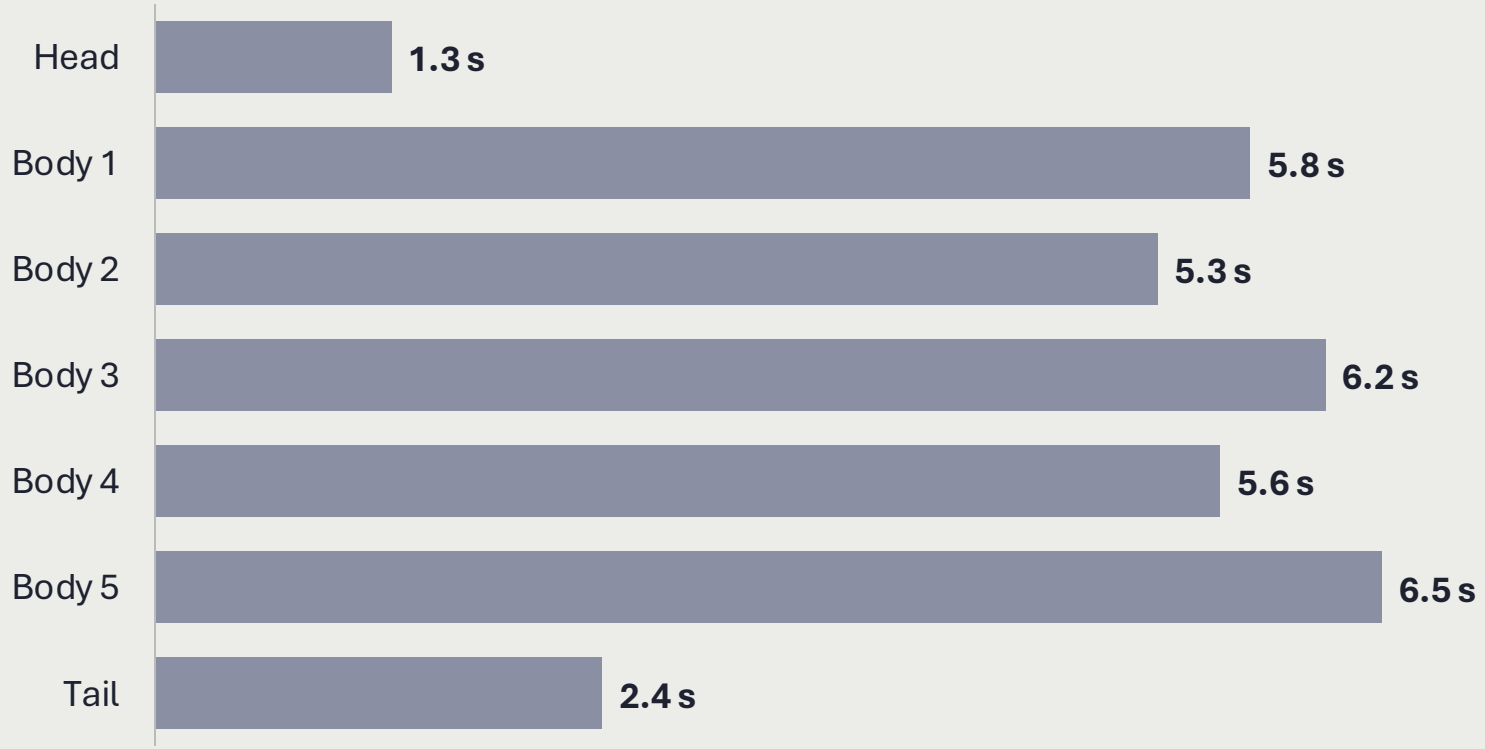
0% — 100%

Reducers: 99.7% of contributions

Medians and means of ~3-hour interval values from the live Pluralis-8B run. Source: agora.pluralis.ai telemetry API; 40 s window from the paper's config.

Rosters formed in about 5 s, well inside the 40 s matchmaking window

Median roster formation time by stage



5 s

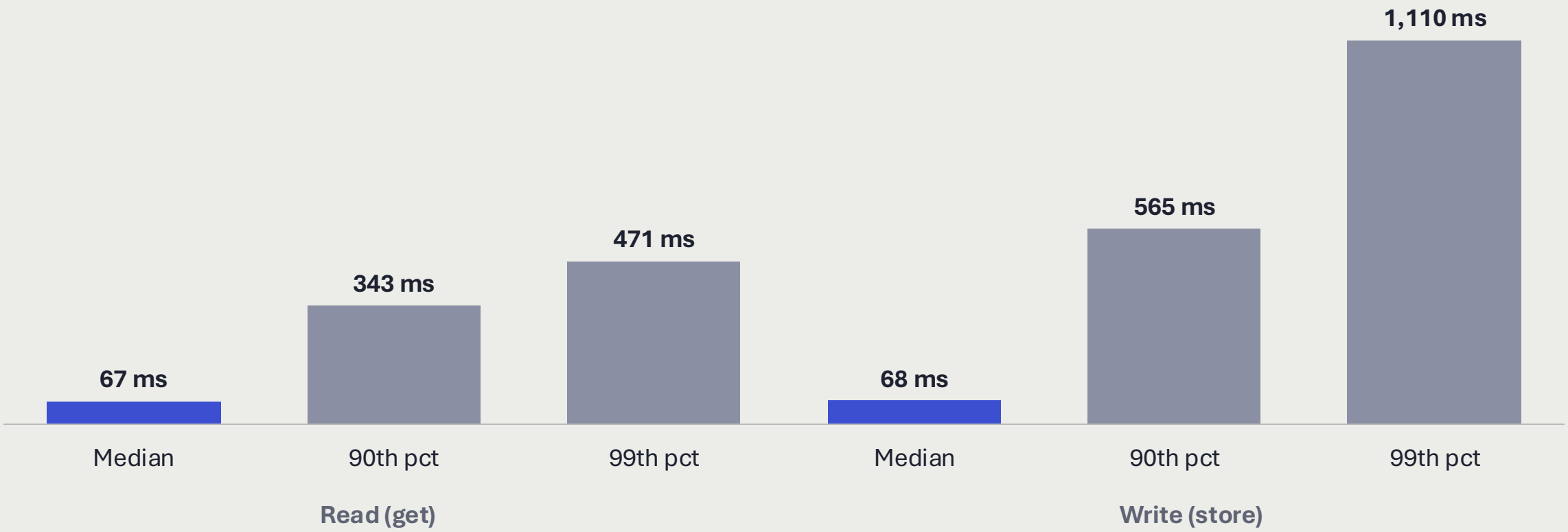
median roster time



Typical roster: 16 peers

Medians of ~3-hour exported interval values, not individual rounds. Source: agora.pluralis.ai telemetry API; Agora config (40 s window)

DHT operation latency

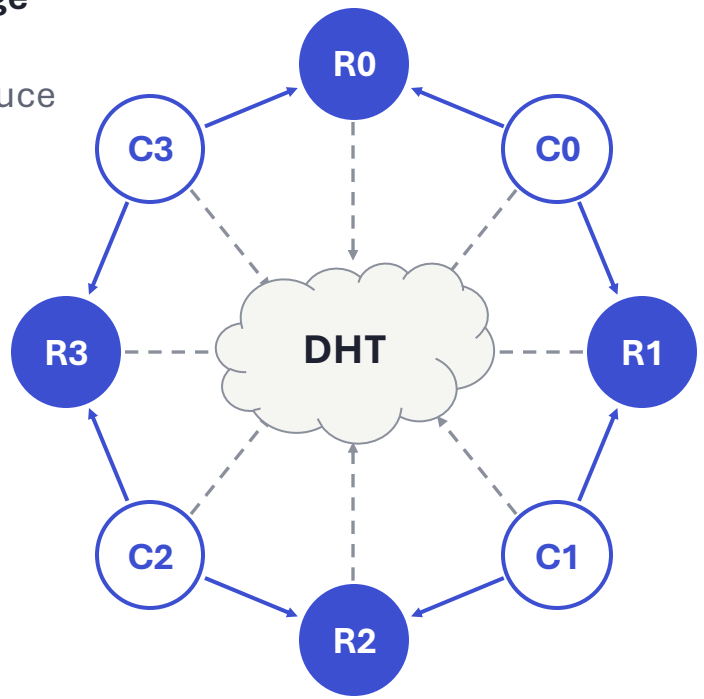


Percentiles across ~3-hour interval averages per worker, not individual calls. Source: agora.pluralis.ai telemetry API

Simulation results

One laptop · one stage

Native Agora DHT, matchmaking, all-reduce



Contributor

Advertisement

Averaging

Reducer

Result	Seed 8101	Seed 8102
Verified rounds	**5/5**	**5/5**
Matching rosters	**5/5**	**5/5**
Ready-to-roster	1.087 s	0.952 s
Failed tensor parts	**0**	**0**
Mean loss	0.7953 → 0.00188	1.1724 → 0.00149

Network: injected delay $(RTT_i + RTT_j)/2$ ms per leg, RTTs from the 8B run · 40 s matchmaking · 150 s timeout

Training: 100-value CPU model, synthetic data · 5% of weights averaged every 20 steps · 2 s steps · 5 rounds/seed

Timing means use 40 peer-round observations per seed. Source: baseline8b_pilot_20261013_a trial configuration and verified report